

A Free MATLAB Alternative Called Octave

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Matrix Laboratory (MATLAB) is an all-time favourite tool for electronics engineers working on data interpretation and analysis. There are a lot of reasons that help justify its position—the ease of use, flexibility, documentation and add-ons in the form of tool boxes, among others. But for beginners, enthusiasts and academicians, MATLAB is not always a smart choice. The software along with toolboxes is too costly for them to purchase.

In the search for an effective alternative for MATLAB, we cannot compromise on certain features. First feature is obviously compatibility with MATLAB. If at one point of time, the user wants to switch to MATLAB, he or she should not be forced to re-write the entire code. Second, all

computations in the tool should be based on matrices as in MATLAB. Also, the user should be able to create functions in the program. We can add a lot more to this list. In this article we are introducing GNU Octave, a smart open source tool that has all the features that enable it to be a close alternative to MATLAB.

What is GNU Octave

GNU Octave is a high-level programming tool for complex numerical computations. The interactive command line interface (CLI) provides an excellent mechanism for solving linear and non-linear problems numerically. The latest version 3.8.2 is equipped with a graphical user interface (GUI) along with the traditional CLI.

Octave makes use of an interpreter for compiling and executing Octave instructions at run-time.

It can create, save or print plots, graphs and charts with the help of gnuplot and Grace software present in this month's EFY Plus DVD. The tool provides extensive graphics capabilities for data visualisation and manipulation.

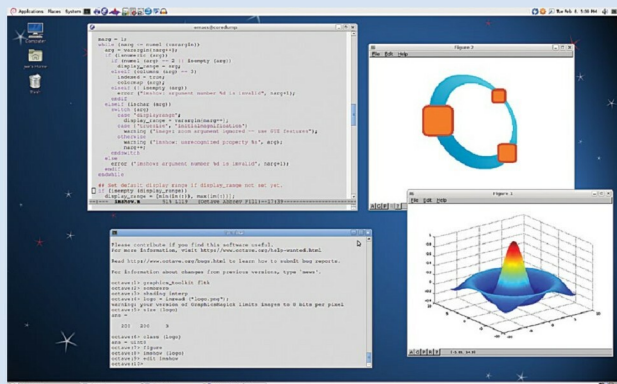
The software is written in C++ using C++ standard library. It is available under the GNU general-public licence and can be used, shared and modified free of cost.

Octave now incorporates GUI

Until recently, the main blame on Octave software was that it did not provide the user with a GUI. The argument the company gave was, if you do not have a pocket calculator, you will be really strong at mathematics. Similarly, if you do not have a GUI, you will learn the language better. But the users were not quite happy

with such an answer. Octave has finally come up with a workable GUI from version 3.8.0 onwards.

Since the GUI is not polished, the graphical interface is not default and the user has to invoke GUI with force-GUI option. Developers are working on the pain areas of the present graphical interface of the software, and will soon come up with another version 4.0.x, with default GUI.



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